

ET LOSSLESS MICROPHONE – CONTINUOUS-DISCRETE BRIDGE

Derived forward from PoDoT = E

The bijection  $\Pi_N(r)=(k,d,\epsilon)$  is algebraic identity

Zero loss. Zero approximation. Zero free parameters.

All files saved to: C:\Users\aezum\Desktop\Civilizational Blueprint\lossless microphone  
Precision: 400 dps (1200 binary digits + guard)  
 $\Lambda = 1200/\ln 2 = 1731.23404906676$

Multiple input devices detected. Showing selection:

| AVAILABLE INPUT DEVICES |                                         |                     |
|-------------------------|-----------------------------------------|---------------------|
| [ 1]                    | Microsoft Sound Mapper - Input          | MME                 |
|                         | 2ch, 44100Hz default                    |                     |
| [ 2]                    | Microphone (HyperX QuadCast S)          | Windows WASAPI      |
|                         | 2ch, 48000Hz default                    |                     |
| [ 3]                    | Primary Sound Capture Driver            | Windows DirectSound |
|                         | 2ch, 44100Hz default                    |                     |
| [ 4]                    | Line In (Realtek HD Audio Line input)   | Windows WDM-KS      |
|                         | 2ch, 44100Hz default                    |                     |
| [ 5]                    | Microphone (Realtek HD Audio Mic input) | Windows WDM-KS      |
|                         | 2ch, 44100Hz default ◀ SELECTED         |                     |
| [ 6]                    | Stereo Mix (Realtek HD Audio Stereo in) | Windows WDM-KS      |
|                         | 2ch, 48000Hz default                    |                     |

Select device number [Enter for auto]: 2  
✓ Selected: Microphone (HyperX QuadCast S) via WASAPI Exclusive  
48000 Hz, 32-bit, 2ch

| HARDWARE – DIRECT ACCESS REPORT |                                               |
|---------------------------------|-----------------------------------------------|
| Device:                         | Microphone (HyperX QuadCast S)                |
| Host API:                       | Windows WASAPI                                |
| Access:                         | WASAPI Exclusive                              |
| Status:                         | ✓ DIRECT – No OS processing – raw ADC samples |
| Channels:                       | 2                                             |
| Native rate:                    | 48000 Hz                                      |
| Bit depth:                      | 32-bit (int32)                                |
| Rates:                          | 48000, 44100, 22050, 16000, 11025             |
| Oversample:                     | 2.4x @20kHz, 12.0x @4kHz                      |

Active device: Microphone (HyperX QuadCast S)

| MAIN MENU                            |
|--------------------------------------|
| 1. Record from microphone            |
| 2. Process a WAV file                |
| 3. Reconstruct from ETLM file        |
| 4. Run verification suite (13 tests) |
| 5. Full demonstration                |
| 6. Hardware capabilities report      |
| 7. Select input device               |
| 8. Exit                              |

Select [1-8]: 6

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Press Enter to return to menu...

Active device: Microphone (HyperX QuadCast S)

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Select [1-8]: 1  
Recording name (Enter for timestamp): 3

ET LOSSLESS MICROPHONE – RECORDING

Direct ADC →  $\Pi_N$  bijection → ETLM (no raw WAV)

| HARDWARE – DIRECT ACCESS REPORT |                                               |
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| Channels:                       | 2                                             |
| Native rate:                    | 48000 Hz                                      |
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Resolution: N=12 | Precision: 400 dps  
Differential tracker:  $d\epsilon = \Lambda \cdot dr/r$  (resolves sub-cell structure)  
 $\sigma_{ET} = 1/\sqrt{12} = 0.28867513$  (structural cell scale)

- RECORDING... press ENTER to stop.
- STOPPED after 9.64s. Captured 459776 frames (9.579s)

Projecting 919552 samples through ET bijection + differential tracker...  
Saving ET lossless → 3.etlm  
Reconstructing → 3.wav (from ETLM)

Differential tracker statistics:  
Total samples: 919552  
Predictions accepted: 16076 (1.819%)  
Measurement overrides: 867506  
Resolved depth: +0.01325 levels  
Total resolved: 32.01 (ADC base: 32)  
 $\sigma_{ET} = 1/\sqrt{12} = 0.28867513$  (structural cell scale)  
 $\partial I$  escalations (F): 82689  
Twilight zone samples: 204854  
Dominant d-family (K): d=12



Lattice analysis:  
Total samples: 919552  
Non-zero samples: 917983  
Zero samples: 1569 (0.1706%)  
k range: [-678, -20] (span=658)  
Dynamic range: 54.83 octaves = 330.1 dB  
ε statistics: mean=-2.27885¢, RMS=25.549¢, range=[-49.9527, 49.991]¢  
Cell transitions: 821572 (89.34% of sample pairs)  
d-family changes: 759935 (82.64%)  
Sign changes: 35874 (3.901%)  
d-family distribution (sublattice families at N=12):  
d= 1: 72426 ( 7.89%)  
d= 2: 80178 ( 8.734%)  
d= 3: 132013 ( 14.38%)  
d= 4: 128451 ( 13.99%)  
d= 5: 4964 ( 0.5408%)  
d= 6: 146293 ( 15.94%)  
d= 10: 16348 ( 1.781%)  
d= 12: 275933 ( 30.06%)  
d= 15: 4333 ( 0.472%)  
d= 20: 9276 ( 1.01%)  
d= 30: 13522 ( 1.473%)  
d= 60: 34246 ( 3.731%)

Output files (in C:\Users\aezum\Desktop\Civilizational Blueprint\lossless microphone):  
3.etlm 34027313 bytes (ET lossless)  
3.wav 3678252 bytes (playback)

The ETLM IS the recording. The WAV is reconstructed from it.  
The lattice is a perfect deterministic crystal. ε is exact.  
Only limitation: engineering (building the hardware).  
[919552 samples stored in session]

Press Enter to return to menu...

Active device: Microphone (HyperX QuadCast S)

MAIN MENU

1. Record from microphone

2. Process a WAV file

3. Reconstruct from ETLM file

4. Run verification suite (13 tests)

5. Full demonstration

6. Hardware capabilities report

7. Select input device

8. Exit

Select [1-8]: 4

ET LOSSLESS MICROPHONE – VERIFICATION SUITE

ET LOSSLESS MICROPHONE – VERIFICATION SUITE  
Testing that the continuous-discrete bridge has ZERO loss

1. Round-Trip (Mathematical Constants)

|                   |    |     |    |      |    |    |     |              |       |   |
|-------------------|----|-----|----|------|----|----|-----|--------------|-------|---|
| π                 | N= | 12: | k= | 20   | d= | 3  | ε=  | -18.2046¢    | EXACT | ✓ |
| e                 | N= | 12: | k= | 17   | d= | 12 | ε=  | 31.234¢      | EXACT | ✓ |
| φ                 | N= | 12: | k= | 8    | d= | 3  | ε=  | 33.0903¢     | EXACT | ✓ |
| √2                | N= | 12: | k= | 6    | d= | 2  | ε=- | 1.9634e-397¢ | EXACT | ✓ |
| 2/3 (Koide)       | N= | 12: | k= | -7   | d= | 12 | ε=  | -1.955¢      | EXACT | ✓ |
| 3/2 (fifth)       | N= | 12: | k= | 7    | d= | 12 | ε=  | 1.955¢       | EXACT | ✓ |
| α <sup>-1</sup>   | N= | 12: | k= | 85   | d= | 12 | ε=  | 18.0934¢     | EXACT | ✓ |
| μ(e/p)            | N= | 12: | k= | 130  | d= | 6  | ε=  | 10.9644¢     | EXACT | ✓ |
| σ_ET              | N= | 12: | k= | -22  | d= | 6  | ε=  | 49.0225¢     | EXACT | ✓ |
| Λ                 | N= | 12: | k= | 129  | d= | 4  | ε=  | 9.10208¢     | EXACT | ✓ |
| 10 <sup>-10</sup> | N= | 12: | k= | -399 | d= | 4  | ε=  | 36.8629¢     | EXACT | ✓ |
| 10 <sup>10</sup>  | N= | 12: | k= | 399  | d= | 4  | ε=  | -36.8629¢    | EXACT | ✓ |

→ ✓ PASS

2. Round-Trip (Audio Sample Values)

|                  |         |   |
|------------------|---------|---|
| 0.5              | → EXACT | ✓ |
| -0.5             | → EXACT | ✓ |
| 0.001            | → EXACT | ✓ |
| -0.001           | → EXACT | ✓ |
| 0.999999         | → EXACT | ✓ |
| -0.999999        | → EXACT | ✓ |
| 0.00001          | → EXACT | ✓ |
| 1/8388608        | → EXACT | ✓ |
| -1/8388608       | → EXACT | ✓ |
| 0.7071067811865  | → EXACT | ✓ |
| 123456/8388608   | → EXACT | ✓ |
| -7654321/8388608 | → EXACT | ✓ |

→ ✓ PASS

3. WAV Round-Trip (File I/O)

|         |                              |   |
|---------|------------------------------|---|
| sine    | : 2400 samples, 0 mismatches | ✓ |
| chirp   | : 2400 samples, 0 mismatches | ✓ |
| impulse | : 2400 samples, 0 mismatches | ✓ |
| dc      | : 2400 samples, 0 mismatches | ✓ |
| square  | : 2400 samples, 0 mismatches | ✓ |

→ ✓ PASS

4. Cross-Resolution Transition

20 cross-resolution transitions tested  
All transitions match direct projection: ✓

→ ✓ PASS

5. Differential Control Identity

|                |                        |                     |                |   |
|----------------|------------------------|---------------------|----------------|---|
| r=3.1415926535 | dε/dr: exact=551.06891 | numeric=551.06891   | err=1.592e-201 | ✓ |
| r=2.7182818284 | dε/dr: exact=636.88541 | numeric=636.88541   | err=1.839e-201 | ✓ |
| r=0.5          | dε/dr: exact=3462.4681 | numeric=3462.4681   | err=1.0e-200   | ✓ |
| r=2.0          | dε/dr: exact=865.61702 | numeric=865.61702   | err=2.5e-201   | ✓ |
| r=137.036      | dε/dr: exact=12.633425 | numeric=12.633425   | err=3.649e-203 | ✓ |
| Λ = 1200/ln2   | = 1731.23404906676     | (verified constant) |                |   |

→ ✓ PASS

6. Lattice Arithmetic

|                               |       |   |                |       |   |
|-------------------------------|-------|---|----------------|-------|---|
| 3.141592×2.718281: k_x=37 κ=0 | EXACT | ✓ | ÷: k_÷=3 κ=0   | EXACT | ✓ |
| 0.666666×1.5: k_x=0 κ=0       | EXACT | ✓ | ÷: k_÷=-14 κ=0 | EXACT | ✓ |
| 137.036×1836.152: k_x=215 κ=0 | EXACT | ✓ | ÷: k_÷=-45 κ=0 | EXACT | ✓ |

→ ✓ PASS



7. Precision Scaling

The bijection is ALGEBRAIC IDENTITY – zero mathematical error.  
Guard digits (100) absorb transcendental residual.  
At each dps, truncated comparison → EXACT 0.

dps= 100 (internal=200): EXACT 0 – lossless ✓  
dps= 200 (internal=300): EXACT 0 – lossless ✓  
dps= 400 (internal=500): EXACT 0 – lossless ✓

All levels EXACT: the bijection has ZERO error.  
This is not approximation. This is algebraic identity.

→ ✓ PASS

8. Bipolar Sign Encoding

0.5 : sign=+1 k\_0=0 d\_0=1 EXACT ✓  
-0.5 : sign=-1 k\_0=6 d\_0=2 EXACT ✓  
0.123456 : sign=+1 k\_0=0 d\_0=1 EXACT ✓  
-0.123456 : sign=-1 k\_0=6 d\_0=2 EXACT ✓  
0.999 : sign=+1 k\_0=0 d\_0=1 EXACT ✓  
-0.999 : sign=-1 k\_0=6 d\_0=2 EXACT ✓  
0.001 : sign=+1 k\_0=0 d\_0=1 EXACT ✓  
-0.001 : sign=-1 k\_0=6 d\_0=2 EXACT ✓

→ ✓ PASS

9. Zero Sample Handling

Zero sample: is\_zero=True, sign=0, reconstructed=0.0 ✓

→ ✓ PASS

10. Tower Escalation

π:  
N= 12: k= 20 d= 3 transition=✓  
N= 60: k= 99 d= 20 transition=✓  
N= 420: k= 694 d= 210 transition=✓  
N= 840: k= 1387 d= 840 transition=✓  
N= 2520: k= 4162 d= 1260 transition=✓  
N=27720: k= 45779 d=27720 transition=✓  
φ:  
N= 12: k= 8 d= 3 transition=✓  
N= 60: k= 42 d= 10 transition=✓  
N= 420: k= 292 d= 105 transition=✓  
N= 840: k= 583 d= 840 transition=✓  
N= 2520: k= 1749 d= 840 transition=✓  
N=27720: k= 19244 d= 6930 transition=✓  
σ\_ET:  
N= 12: k= -22 d= 6 transition=✓  
N= 60: k= -108 d= 5 transition=✓  
N= 420: k= -753 d= 140 transition=✓  
N= 840: k= -1506 d= 140 transition=✓  
N= 2520: k= -4517 d= 2520 transition=✓  
N=27720: k= -49688 d= 3465 transition=✓  
0.5:  
N= 12: k= -12 d= 1 transition=✓  
N= 60: k= -60 d= 1 transition=✓  
N= 420: k= -420 d= 1 transition=✓  
N= 840: k= -840 d= 1 transition=✓  
N= 2520: k= -2520 d= 1 transition=✓  
N=27720: k= -27720 d= 1 transition=✓

→ ✓ PASS

11. Backbone Bridge (Identity G)

π : Cont\_EML=19.817954 → T\_round k=20 δ=-0.182046 → Webb d=3 ✓  
e : Cont\_EML=17.31234 → T\_round k=17 δ=0.31234 → Webb d=12 ✓  
φ : Cont\_EML=8.330903 → T\_round k=8 δ=0.330903 → Webb d=3 ✓  
2/3 : Cont\_EML=-7.01955 → T\_round k=-7 δ=-0.01955 → Webb d=12 ✓  
α<sup>-1</sup> : Cont\_EML=85.180934 → T\_round k=85 δ=0.180934 → Webb d=12 ✓  
σ\_ET : Cont\_EML=-21.509775 → T\_round k=-22 δ=0.490225 → Webb d=6 ✓  
Backbone factorization: Π\_N = Disc\_Webb ◦ T\_round ◦ Cont\_EML  
Three backbones converge at N=12 (Theorem 15.15)

→ ✓ PASS

12. d-Family Composition (Identity C)

C.3 Residue symmetry Res(d) = {k, N-k}: ✓  
C.4 Universal d=1 channel (1 ∈ d⊗d ∀d): ✓  
C.5 d=12 universality (12⊗12 = all): [1, 2, 3, 4, 6, 12] ✓  
C.6 Lattice multiplication matches direct: ✓

→ ✓ PASS

13. ∂I Boundary (Identity F)

F.1 Half-cell width = 600/N = 50.0¢: ✓  
F.2 Adjacent cells always differ in d-family: ✓ (k=-24..24)  
F.3 Boundary detection for cell-edge values: ✓

→ ✓ PASS

VERIFICATION SUMMARY

- ✓ 1. Round-Trip (Mathematical Constants)
- ✓ 2. Round-Trip (Audio Sample Values)
- ✓ 3. WAV Round-Trip (File I/O)
- ✓ 4. Cross-Resolution Transition
- ✓ 5. Differential Control Identity
- ✓ 6. Lattice Arithmetic
- ✓ 7. Precision Scaling
- ✓ 8. Bipolar Sign Encoding
- ✓ 9. Zero Sample Handling
- ✓ 10. Tower Escalation
- ✓ 11. Backbone Bridge (Identity G)
- ✓ 12. d-Family Composition (Identity C)
- ✓ 13. ∂I Boundary (Identity F)

OVERALL: ALL PASS ✓ – LOSSLESS VERIFIED

Press Enter to return to menu...